

Bitcoin Follow-the-Money Trace

Interim evidence package - live target chain data not retrieved in this execution session

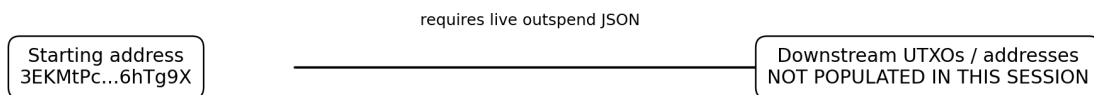
Target Bitcoin address: 3EKMtPc89gGsJHaCtcbWyKT4zRLH6hTg9X

Requested focus: funds present at some point during calendar year 2020 (UTC)

Package creation date: 18 August 2026

Evidence status

The address itself was validated, and the public explorer API methods needed for the trace were verified. However, the target-specific transaction JSON could not be retrieved inside this ChatGPT execution environment. Consequently, this interim report does **not** state any transaction IDs, downstream addresses, or balances that were not actually obtained. The bundled tracer can populate the complete report from the public Bitcoin chain on an internet-connected machine.



This is a retrieval-status diagram, not a claim about where the BTC went.
Run trace_bitcoin.py to generate the evidence-based flow graph.

The diagram above deliberately shows only the retrieval state. It is not a substitute for the requested money-flow graph and makes no downstream claim.

1. Starting address validation

Field	Validated value
Address	3EKMTpC89gGsJHaCtcbWykT4zRLH6hTg9X
Network	Bitcoin mainnet
Encoding	Base58Check
Version byte	0x05
Type	P2SH (Pay-to-Script-Hash)
Hash160	8a8125f0674cc2d3d0e4ac33f14bd23ecdf85b6c
Checksum	a11aba9a - valid

No current balance is shown here because the live address summary was not successfully retrieved. Leaving this blank is intentional.

2. How the live trace is defined

The bundled tracer first finds every output paid to the target address that existed at any point between 2020-01-01 00:00:00 UTC and 2020-12-31 23:59:59 UTC. An output funded before 2020 is included if it had not yet been spent when 2020 began. An output funded during 2020 is included if it existed for any portion of that year.

Each selected output is then followed by its exact outpoint (transaction ID plus output index) to the transaction that spends it. This is stronger than simply drawing address-to-address arrows because the Bitcoin ledger directly records which previous output a transaction input consumes.

When a downstream transaction combines tracked value with unrelated inputs, the blockchain does not encode which original satoshis correspond to which new outputs. From that point the tracer switches to proportional (haircut) attribution and labels the result as a heuristic rather than an exact provenance statement.

3. Files produced by the live tracer

File	Purpose
bitcoin_follow_the_money_report.pdf	Populated narrative report with graph, relevant 2020 UTXOs, downstream addresses and current whole-address balances.
trace_graph.png	Readable static overview of the largest/most relevant portion of the flow.
trace_graph.graphml	Full graph for tools such as Gephi; not limited to the static-image node cap.
starting_utxos.csv	Exact starting outpoints that existed during the requested period.
transactions.csv	Spending transactions, fees, tracked-input ratio, and evidence method.
flow_edges.csv	Address/transaction/output graph edges with output value and attributed value.
terminal_outputs.csv	Where a branch stopped: unspent, hop cap, small-value threshold, or high fan-out.
addresses.csv	Current confirmed and mempool-inclusive balance of encountered addresses; these are whole-address balances.
raw/*.json	Raw API responses retained so the derived report can be independently audited.

4. Balance interpretation

A current downstream address balance is the balance of that entire Bitcoin address at retrieval time. It is not the amount that can still be attributed to the starting address. If an address later received unrelated funds, its whole-address balance can be much larger than the tracked lineage. The CSV therefore keeps balance fields separate from attributed-flow fields.

5. Default trace limits

To keep a forensic graph interpretable, the script defaults to four downstream hops, a 0.00001 BTC minimum attributed amount, a stop at transactions with more than 30 outputs, and up to 250 current address-balance lookups. All limits are command-line options and can be increased.

6. Reproduction instructions

From this ZIP, on a normal internet-connected machine:

```
python -m pip install -r requirements.txt
```

```
python trace_bitcoin.py --address 3EKMtPc89gGsJHaCtcbWykT4zRLH6hTg9X --from-date 2020-01-01 --to-date 2020-12-31 --out live_trace
```

The program uses the public mempool.space Esplora-compatible REST API by default. The API base can be changed with `--api-base` to another compatible Esplora endpoint.

7. Public API references

Mempool REST API documentation: <https://mempool.space/docs/api/rest>

Blockstream Esplora API specification: <https://github.com/Blockstream/esplora/blob/master/API.md>

Blockchain.com Data API documentation: https://www.blockchain.com/explorer/api/blockchain_api

8. Privacy and attribution limits

This package intentionally excludes personal identifiers supplied outside the blockchain. Such identifiers are not needed to trace public UTXO spends. It also does not assign a downstream address to a person or organization unless a separate, reliable attribution source supports that claim.

Bitcoin transaction analysis can show ledger relationships. It cannot, by itself, prove beneficial ownership, intent, or legal entitlement.